ASEAN
GEOAI FUSION 2026

Module 4:
Foundational Knowledge: Foundation in GEOAI (FD1004)

AGAIF 2026 Boot Camp 1
Level: Foundation of GeoAI

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From Connectivity Gaps to Evidence-Based Decisions

The learning journey across Sessions 4–6

SESSION 4

UNDERSERVED COMMUNITIES' IDENTIFICATION

UNDERSTAND & IDENTIFY

Who may be underserved — and why?

Combine telecommunications conditions with population, community and spatial context.

Telecom Network · Population · Communities · Public Services

Potential underserved communities

SESSION 5

CASE STUDY – ASEAN TELECOMMUNICATIONS CHALLENGE

COMPARE & REASON

When connectivity needs compete, what evidence should guide priority?

Examine ASEAN telecommunications challenges, evidence layers and competing planning considerations.

Impact · Urgency · Feasibility · Public Value

Transparent priority reasoning

SESSION 6

HANDS-ON

APPLY & RECOMMEND

How can spatial evidence be transformed into a defensible recommendation?

Apply the guided spatial screening workflow, interpret the evidence and extend it into priority and intervention recommendations.

Jupyter Workflow · Spatial Screening · Candidate Areas · Recommendation

Evidence-based decision support

Why This Module Matters

Connectivity inequality is not only a network problem. It is a spatial problem involving people, place, public services and planning priorities.

GeoAI helps connect these evidence layers so that connectivity decisions can be explained and defended.

IDENTIFY → **COMPARE** → **APPLY** → **RECOMMEND**

Learning progression: Understand the problem → Reason with evidence → Apply the workflow → Support a decision



Session 4:
Underserved Communities' Identification

GeoAI for Underserved Connectivity Intelligence

Objectives — From Data Layers to Intervention Priority

SESSION OBJECTIVES

01 Reframe underserved connectivity as a **multi-dimensional spatial intelligence problem**, not only a coverage gap.

02 Understand how network performance, population exposure, socio-economic vulnerability, public facilities, infrastructure proximity, and terrain/accessibility can be fused into one **analytical framework**.

03 Apply GeoAI methods such as grid-based scoring, hotspot analysis, predictive modelling, graph analysis, and uncertainty mapping to support **intervention prioritisation**.

04 Translate underserved-area analysis into policy-relevant outputs: priority maps, ranked intervention zones, recommended infrastructure actions, and uncertainty notes.

By the end of this session

> Participants should be able to **identify potential underserved communities** by combining telecommunications and geospatial evidence and **explain the spatial evidence** supporting the identification.

Key takeaway:
This session moves from data layers to explainable intervention priorities.

Problem framing → Data fusion → GeoAI methods → Policy outputs

GeoAI for Underserved Connectivity Intelligence

Reframing the Problem: From Coverage Gap to Connectivity Inequality



> Underserved identification is not a simple no-signal problem. It is a **multi-dimensional decision problem** involving technical service quality, population exposure, socio-economic vulnerability, public service impact, and deployment feasibility.

KEY DIMENSIONS

Coverage Availability	Is mobile broadband technically available?
Service Quality	Are speed, latency, and reliability sufficient?
Population Exposure	How many people are affected?
Socio-economic Vulnerability	Are affected users rural, low-income, or isolated?
Public Service Impact	Are schools, clinics, or critical services affected?
Implementation Feasibility	What intervention is realistic: tower, fibre, backhaul, satellite, or sharing?

Key takeaway: We are not just mapping weak coverage — we are prioritising connectivity inequality for evidence-based intervention.

GeoAI for Underserved Connectivity Intelligence

Conceptual Model: Underserved Area as a Multi-Dimensional Risk Surface

Underserved Priority Score (UPS)

$$UPS_i = w_1\, NQD_i + w_2\, PE_i + w_3\, SV_i + w_4\, PFC_i + w_5\, IDC_i + w_6\, AC_i$$

Underserved areas are modelled as a weighted spatial risk surface that integrates network quality deficit, population exposure, socio-economic vulnerability, public facility criticality, infrastructure distance constraints, and accessibility constraints.

UPS COMPONENTS

NQD — Network Quality Deficit	Weak speed, high latency, poor reliability
PE — Population Exposure	How many people are affected?
SV — Socio-economic Vulnerability	Rurality, poverty, isolation, low adoption risk
PFC — Public Facility Criticality	Schools, clinics, emergency and public services
IDC — Infrastructure Distance Constraint	Distance to tower, fibre, road, or backhaul
AC — Accessibility Constraint	Terrain, remoteness, and deployment difficulty

Key takeaway: UPS converts multiple spatial layers into a transparent and adjustable intervention priority model.

GeoAI for Underserved Connectivity Intelligence

Data Fusion Architecture for ASEAN-Level Underserved Mapping

DATA FUSION ARCHITECTURE

- **Network Performance Data**
  speed, latency, reliability, signal quality
- **Population Grid**
  population density, settlement exposure
- **Infrastructure Layer**
  cell towers, fibre, backhaul, roads
- **Public Facilities**
  schools, clinics, emergency services
- **Socio-economic Indicators**
  poverty, rurality, digital inclusion risk
- **Terrain & Accessibility**
  slope, land cover, remoteness

→ **GeoAI Integration & Spatial Feature Engineering**
harmonisation, gridding, feature extraction, proximity analysis

→ **Underserved Area Intelligence Output**
priority map, ranked intervention zones, evidence-based decision support

EXAMPLE DATA SOURCES

Ookla Open Data	mobile/fixed speed and latency surfaces
WorldPop	high-resolution population exposure estimation
OpenCelliD	open cell tower and base station reference
OpenStreetMap	roads, buildings, facilities, and accessibility context
HDX / Admin Boundaries	district, province, and regional aggregation
National / Operator / Regulator Data	ground-truth network and infrastructure detail

Key takeaway: High-value underserved mapping depends on fusing technical, demographic, infrastructure, and contextual spatial data.

GeoAI for Underserved Connectivity Intelligence

GeoAI Methods: Beyond Overlay Mapping

FROM MAPS TO INSIGHTS
Transform spatial data into intelligent decisions for connectivity equity.

KEY METHODS

01 SPATIAL CLUSTERING
Identify clusters of low-speed or poor-quality service areas using algorithms such as DBSCAN or K-Means.

02 HOTSPOT ANALYSIS
Detect statistically significant hotspots of weak connectivity (e.g., Getis-Ord Gi*, Moran's I) to prioritise critical zones.

03 GRID-BASED SCORING
Standardise diverse datasets into uniform grid cells and compute multi-criteria scores for fair comparison.

04 PREDICTIVE MODELLING
Use ML models (RF, XGBoost, LightGBM) to predict underserved probability using network, demographic and terrain features.

05 SPATIAL INTERPOLATION
Estimate network performance in unmeasured areas using kriging, IDW or other spatial interpolation methods.

06 GRAPH ANALYSIS
Assess distance and connectivity to infrastructure (e.g., towers, fiber nodes, roads) using network analysis on graph models.

07 MULTI-CRITERIA DECISION ANALYSIS (MCDA)
Combine multiple weighted criteria transparently to rank intervention priority areas.

08 UNCERTAINTY MAPPING
Quantify confidence of predictions and highlight areas with sparse data or high variability for informed decision-making.

The goal is not just to map problems, **but to intelligently prioritise solutions.**

Evidence-based prioritisation | Targeted intervention | Cost-efficient deployment

Sources: ITU UMC Framework (2023) | GSMA SOMIC 2025 | OECD (2023) | Ookla Open Data | WorldPop | OpenStreetMap | OpenCellID

GeoAI for Underserved Connectivity Intelligence

Designing an Underserved Priority Index

UNDERSERVED PRIORITY INDEX

Dimension	Possible Indicator	Interpretation
Network Quality Deficit	Low download speed, high latency, weak signal	Technical service gap
Population Exposure	Population within weak-service grid	Number of affected users
Social Vulnerability	Rurality, poverty, low digital adoption proxy	Risk of exclusion
Critical Facilities	Schools, clinics, emergency centres affected	Public service impact
Infrastructure Gap	Distance from tower, fibre, road, backhaul	Deployment challenge
Strategic Value	Border area, island, disaster-prone zone, economic corridor	Policy relevance

DECISION LOGIC

> The best intervention area is not always the weakest-signal area. It is the area where weak connectivity, high population exposure, social need, and feasible intervention overlap.



Key takeaway: The index should rank areas based on combined technical, social, and strategic need — not signal weakness alone.

Transparent ranking | Policy relevance | Targeted intervention | Impact-driven planning

Sources: OECD (2023) | World Bank (2023) | ASEAN Digital Masterplan 2025 | ITU UMC Framework

GeoAI for Underserved Connectivity Intelligence

ASEAN Use Case: Cross-Border Digital Inclusion Challenge

> ASEAN is not homogeneous.
> The best solutions adapt across urban centres, rural interiors, islands, and border communities.



ASEAN CONNECTIVITY CHALLENGE SCENARIOS

Region Type	Challenge
Dense Urban Areas	Congestion, quality degradation, uneven service
Rural Interiors	Sparse infrastructure, difficult terrain, low commercial return
Islands / Coastal Areas	Backhaul limitations, disaster vulnerability
Border Communities	Administrative complexity, roaming, uneven investment
Economic Corridors	High demand, infrastructure pressure
Education / Health Priority Zones	Service quality needed for learning and telemedicine

EXPECTED PARTICIPANT OUTPUTS

01 Underserved Priority Map

02 Top 10 priority intervention grids / districts

03 Rationale for ranking

04 Recommended intervention type: tower, small cell, fibre extension, backhaul upgrade, satellite, infrastructure sharing

05 Uncertainty / data limitation note

Key takeaway: Strong teams will not only map underserved areas — they will justify policy-relevant, context-aware intervention decisions.

Regional adaptability | Policy relevance | Intervention strategy | Evidence-based decisions

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Sources: ASEAN Digital Masterplan 2025 | ASEAN Digital Masterplan 2030 | ITU UMC Framework | GSMA SOMIC 2025

GeoAI for Underserved Connectivity Intelligence

Real ASEAN Connectivity Use Cases

> These are not just telecom infrastructure projects. They are real regional examples of how ASEAN countries must decide where to invest first, who is most affected, which technology is most suitable, and how to justify intervention using evidence.

Malaysia

JENDELA: National Digital Infrastructure Plan



National plan to improve broadband coverage, quality, and 4G readiness, especially for rural and underserved communities including Sabah and Sarawak.

Challenge / demand: uneven broadband quality; online learning, remote work, digital services, and rural connectivity needs.

Source: Malaysia.gov.my / JENDELA

Indonesia

SATRIA-1 for Remote Public Facilities



Satellite connectivity for schools, hospitals, health centres, village offices, and security services in 3T frontier, outermost, and disadvantaged regions.

Challenge / demand: archipelagic geography, remote islands, expensive fibre rollout, and public service access.

Source: SATRIA-1 / ACN Newswire

Thailand

Net Pracharat Rural Broadband



High-speed internet and community Wi-Fi expanded to rural villages to support digital inclusion and local socio-economic development.

Challenge / demand: rural-urban digital gap, local business, education, e-government, and community access.

Source: Net Pracharat

Philippines

Free Wi-Fi for All / National Broadband Program



Public Wi-Fi and broadband expansion targeted geographically isolated and disadvantaged areas, including remote island communities and public facilities.

Challenge / demand: island isolation, public internet access, disaster-prone areas, schools, clinics, and transport hubs.

Source: PNA / UN SDG Action

Vietnam

Broadband Expansion to Remote and Mountainous Areas



Broadband rollout prioritised remote villages, border regions, islands, ethnic minority communities, schools, and health stations.

Challenge / demand: mountainous terrain, remaining uncovered settlements, vulnerable communities, and access to digital services.

Source: Vietnam News

Key takeaway: Real ASEAN cases show that intervention priorities depend on geography, population need, public service impact, and technology feasibility.

Regional diversity | Investment priority | Technology choice | Evidence-based action

GeoAI for Underserved Connectivity Intelligence

Policy Implication Visualisation

> Policy visualisation is not just about showing maps; it is about making intervention priorities interpretable, defensible, and actionable.

GEOAI OUTPUTS

Output	Explanation
Underserved Priority Map	identifies high-priority underserved zones.
Population Exposure Map	shows how many people are affected.
Critical Facilities Overlay	reveals exposure of schools, clinics, and emergency services.
Ranked Intervention Zones	prioritises districts / grids for action.
Uncertainty Layer	highlights where data is sparse or predictions are less certain.

DECISION QUESTIONS

- Which areas should be prioritised first?
- Who is most affected?
- What intervention type is most suitable?
- Where is additional data collection needed?

POLICY & PLANNING ACTIONS

- **Target universal service funding**
  direct resources to high-impact underserved areas.
- **Prioritise infrastructure rollout**
  focus on highest-need districts and grids.
- **Support shared infrastructure strategy**
  reduce duplication and lower deployment costs.
- **Strengthen rural, island, and border connectivity planning**
  address geographic and access challenges.
- **Enable monitoring and evaluation dashboards**
  track progress and adjust policies in real time.

Key takeaway: Effective visualisation transforms spatial analysis into transparent and actionable policy decisions.

Decision support | Investment priority | Intervention planning | Digital inclusion

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Sources: ASEAN Digital Masterplan 2025/2030 | ITU UMC Framework | GSMA SOMIC 2025 | OECD (2023)

GeoAI for Underserved Connectivity Intelligence

Conclusion — From Underserved Mapping to Evidence-Based Connectivity Intervention

KEY INSIGHTS

	Insight	Meaning
01	Coverage alone is insufficient	Underserved communities may face weak quality, poor affordability, low adoption, or limited access to devices and services.
02	Data fusion creates intelligence	Combining network, demographic, facility, terrain, and infrastructure layers reveals who is affected and why.
03	Priority scoring supports transparency	A clear index helps justify why one location should be prioritised over another.
04	ASEAN requires adaptable models	Urban, rural, island, border, and disaster-prone areas require different intervention strategies.
05	Policy value matters	The final output should support investment, regulation, universal service planning, and digital inclusion.

FINAL TAKEAWAY

> "The goal is not only to find underserved areas, but to justify where, why, and how connectivity intervention should happen."

Transition to Session 5

This framework will be applied in the ASEAN Telecommunications Challenge case study.

Map underserved areas → Rank intervention priorities → Recommend connectivity actions

Key takeaway:
GeoAI transforms underserved-area mapping into explainable, policy-relevant connectivity intervention decisions.

Decision support | Intervention priority | Regional adaptability | Digital inclusion

Sources: ITU UMC Framework | GSMA SOMIC 2025 | ASEAN Digital Masterplan 2025/2030 | OECD (2023)

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THANK YOU

Bootcamp and Hackathon for ASEAN Geospatial Innovation 2026